Metrics across the lifecycle: a framework for evaluating NIH biomedical data ecosystems

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Abstract

The U.S. National Institutes of Health has invested billions of dollars in large-scale biomedical data resources, yet evaluation of these investments has historically been fragmented — each program tracks different metrics in different ways, preventing portfolio-level comparison and informed decision-making. We construct a structured approach built on three complementary prioritization frameworks: *public value* (user engagement, citations, funded grants), *scientific data quality* (metadata completeness, FAIR compliance, data dictionaries), and *operations and finance* (infrastructure reliability, cost sustainability). Critically, we note that the relative importance of each framework shifts predictably across a data resource's lifecycle — from early infrastructure-building, through active use and impact, to long-term sustainability. Drawing on data from the NIH Common Fund Data Ecosystem (CFDE), we identify gaps in current measurement practice and describe the ICC's ongoing effort to close them through a centralized dashboard that integrates publication, citation, GitHub, and web analytics data across the portfolio.

Highlights

- NIH's federated biomedical data ecosystems track different metrics in different ways, which blocks portfolio-level comparison and evidence-driven funding decisions.
- Three prioritization frameworks long familiar in product strategy (public value, scientific quality, operations and finance) carry over cleanly to research data resources.
- Which framework dominates depends on the resource's lifecycle phase; uniform metric panels across phases misallocate funding.
- Fewer than 30% of secondary analyses formally cite their source data, so raw citation counts systematically understate reuse.
- A portfolio-wide dashboard at the NIH Common Fund Data Ecosystem now operationalizes the model across 19 programs in near real time.

Significance

Public data ecosystems are central infrastructure across biology, medicine, and the data sciences, but whether any given resource is succeeding mostly goes unanswered. Each funded program tracks its own metrics in its own way, which makes portfolio-level comparison effectively impossible.

Three prioritization frameworks long familiar in product strategy (public value, scientific quality, operations and finance) carry over cleanly to research data resources. Their relative weight shifts predictably across a resource's lifecycle. Uniform metric panels ignore that shift and misallocate funding as a result.

We ground the argument in the NIH Common Fund Data Ecosystem, where a portfolio-wide dashboard now operationalizes the model across 19 programs. The framework generalizes to every federated biomedical data ecosystem and gives evaluators a shared vocabulary, instead of forcing each program to invent its own.

Public biomedical data resources have become central infrastructure for biology, medicine, and the data sciences. NIH alone has invested billions over the past decade in initiatives that include The Cancer Genome Atlas [1], the Genotype-Tissue Expression project [2], the All of Us Research Program [3], and the 19 programs of the Common Fund. Whether any given resource is succeeding goes mostly unanswered. Each funded program tracks its own metrics in its own way, which blocks portfolio-level comparison and the evidence-driven funding decisions that comparison enables.

We argue that three prioritization frameworks long familiar in product strategy — *public value*, *scientific quality*, and *operations and finance* — map cleanly onto research data resources. Their relative weight shifts predictably across a resource's lifecycle. Treating the frameworks as uniformly important across all phases is the reason portfolio decisions go wrong: early-phase investment in scientific quality looks expensive only when the comparison metric is mid-phase public value.

The NIH Common Fund Data Ecosystem (CFDE) [4] is a cross-cutting initiative that harmonizes metadata and tooling across the 19 Common Fund programs. The Council of Councils Working Group identified "facilitation of new scientific discovery" as the chief metric of CFDE success [5]. We use the CFDE as motivating case study and operational testbed, and describe the Integration and Coordination Center's (ICC) first portfolio-wide dashboard, which operationalizes the model in near real time.

0.0.1 Three frameworks, one lifecycle

The frameworks overlap and sometimes conflict (more scientifically impactful datasets are typically more expensive to create and maintain), and weighted composite scoring such as *Multi-Criteria Decision Analysis* can balance them when needed [6,7]. Their relative weight shifts predictably across a resource's lifecycle. In the **early term** of infrastructure-building and data collection, *scientific quality* and *operations and finance* dominate. In the **mid term** of active use within a funded period, all three matter, with *public value* taking on increasing importance. In the **late, post-funding term**, *operations and finance* and *public value* become the primary lenses. Embedding this lifecycle logic into evaluation design, rather than applying uniform metrics across all phases, is the key practical contribution we describe.

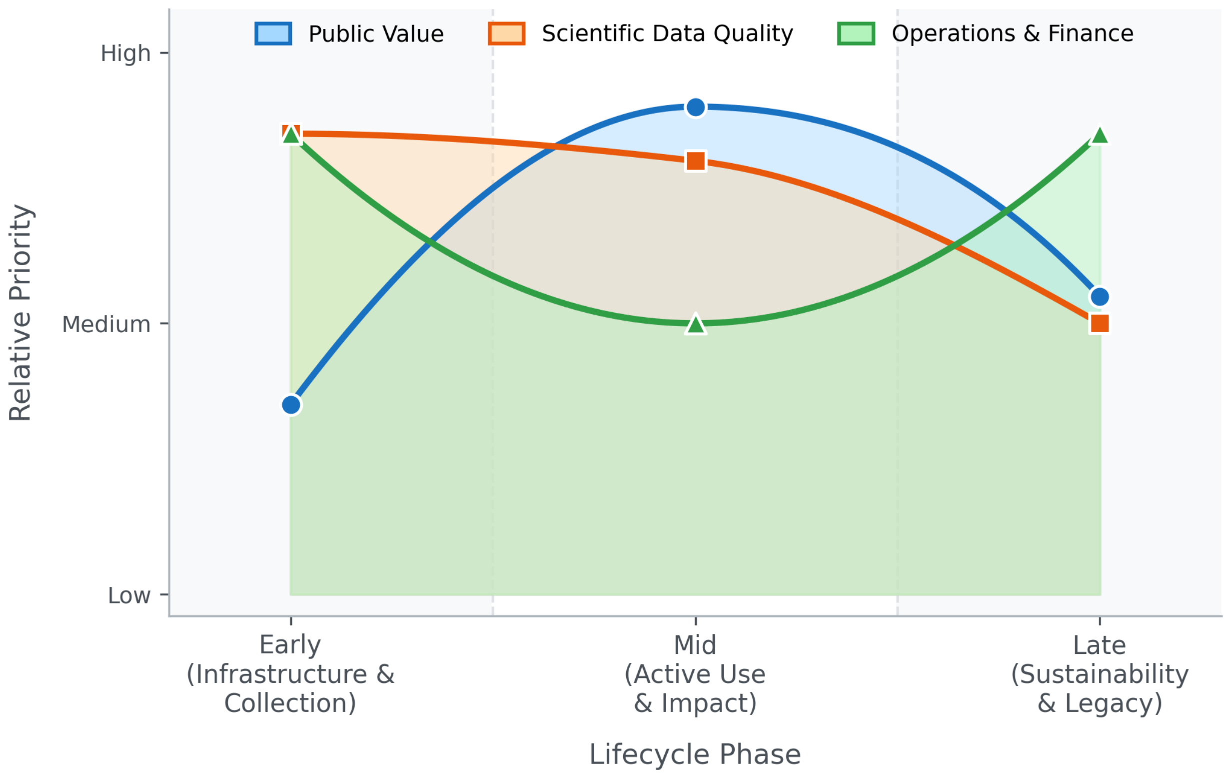

Figure 1: Schematic relative priority of the three metric and evaluation frameworks across the project lifecycle. Lower priority does **not** imply unimportant.

The *public value* framework captures outputs and impacts: user engagement, downloads, funded grants, publications, and downstream scientific influence. The *scientific quality* framework measures reusability and long-term value, guided by the FAIR principles (Findable, Accessible, Interoperable, Reusable) [8,9]. The *operations and finance* framework covers infrastructure reliability and cost sustainability, including robustness as funding sunsets.

0.0.2 Public value

The *public value* framework covers user behavior and scientific impact. Website engagement metrics (*page views*, *time on page*, *actions/triggers*, *new vs. returning users*) are easy to collect via platforms such as Google Analytics or Matomo, but they capture interest rather than actual data use. More direct evidence comes from *downloads* and *compute jobs*; cloud-hosted datasets let analysis activity be measured rather than inferred. For scientific software, the same framing extends to GitHub-style engagement markers such as *stars*, *forks*, and *issues*. Limitations are persistent: bots skew counts, VPN use confounds new-versus-returning tracking, and downloads cannot distinguish active use from passive acquisition. As AI-driven and agent-based traffic on data portals grows, the bot signal will need to be partitioned into noise to remove and demand to interpret.

Citations from PubMed, Scopus, Google Scholar, Clarivate, and OpenAlex give stronger proxies for scientific impact. The original GTEx paper [2] had accumulated 8,630 citations as of January 2026. Composite scores weighted by secondary citations capture cumulative downstream impact that simple counts miss, and citation networks reveal cross-disciplinary collaborations enabled by data sharing; successful sharing also produces a citation boost for the original data generators [10]. Citation counts lag actual use by months to years [11], and journal-scoring alternatives that weight by citing-paper properties better capture subject-specific repository impact than raw counts [12]. The deeper problem is the citation gap: fewer than 30% of secondary analyses formally cite their source data [13]. Standard bibliometrics therefore systematically understate reuse.

Closing the gap likely requires data ecosystems to mine accession numbers (e.g., GEO accession numbers, NCBI BioProject IDs) from publication full text rather than rely on citation databases. A bibliometric analysis of NeuroMorpho.Org found that shared neural morphology data sustained a reuse rate constant for sixteen years and effectively doubled peer-reviewed discoveries in the field [13].

User-centered measures of *perceived value* offer complementary insight. Work integrating the Information System Success Model, the Technology Acceptance Model, and Consumer Perceived Value Theory identifies four entropy-weighted metrics that explain the majority of perceived value: *number of datasets* (breadth of coverage), *data timeliness* (currency for emergent challenges), *search comprehensiveness* (ability to return all relevant records across federated sources), and *system responsiveness* (loading times and API stability) [14]. These can be collected through structured user surveys.

The long-term measure of public value is *practice change* — a shift from hypothesis-driven research toward data-driven discovery, and from evidence-based practice toward practice-based evidence [15]. Concrete indicators include the cloud-versus-local analysis ratio, cross-disciplinary co-authorship across Common Fund programs, and the geographic and institutional breadth of the user base, especially participation by early-career investigators from under-resourced institutions.

0.0.3 Scientific quality

The *scientific quality* framework addresses whether data can be found, understood, and reused. Poor data quality and poor annotation are the leading barriers to reuse [16]. Metadata spans three grain levels: study-level (name, contributors, dates, funding), data-level (storage format, data types, measurement platforms, biospecimen handling and collection conditions), and dataset-level (sample sizes, variable counts, demographic summaries, QC statistics). Summary metrics across these levels — *metadata completeness*, *standardized metadata*, *QC compliance*, *available data dictionary*, *quality data dictionary* — are more tractable for portfolio evaluation than cataloging every study-specific variable. The Standard Evaluation Framework for Large Health Care Data identifies 49 attributes across three domains (collection process, data, and data use) for assessing fitness for purpose [17]. Two of those attributes are particularly load-bearing for federated ecosystems: *representativeness*, the extent to which a dataset generalizes beyond its source population (critical for avoiding both Type 1 and Type 2 errors in scientific conclusions drawn from the data), and *linkability*, the extent to which data integrate with external sources — a paramount concern for an ecosystem whose mission is cross-program query.

For scientific software, quality follows from open-source development practices: *README* and *LICENSE* files, automated testing [18], and inclusion in curated ecosystems like Bioconductor [19]. The ELIXIR "Top 10 Metrics for Life Science Software Good Practices" provides a prioritized scorecard spanning version control, discoverability, automated builds, test data, and documentation [20]. "Minimal Reimplementation" — preferring established libraries over custom code — is particularly relevant for federated ecosystems, where ad hoc code multiplies technical debt across the portfolio. PIsCO (Performance Indicators framework for Collection of bioinformatics resource metrics) automates server-side collection and visualization of these metrics so they can flow directly into evaluation dashboards [20].

A structured assessment of overall quality comes from FAIR maturity models. The shift from first-generation FAIR metrics (manually assessed) to second-generation maturity indicators (automated and machine-measurable) [9] is essential for the CFDE, where the volume of metadata prevents manual audit. The FAIRplus Dataset Maturity (FAIRplus-DSM) model [21], developed for the life sciences, defines six levels: Level 0 (no identifiers, local storage) through Level 5 (provenance in cross-community languages, enterprise-grade lifecycle management), with intermediate levels covering persistent identifiers, indexed metadata, community reporting standards such as MIAME, and linked-data semantic representation. The Research Data Alliance further classifies FAIR indicators as Essential (globally unique identifiers, open protocols), Important (machine-understandable knowledge representation), or Useful (automated agent access), giving evaluators a tiered roadmap. Automated tools such as the FAIR Evaluator test resources at scale via Compliance Tests, making continuous *FAIR compliance* monitoring feasible for large portfolios [9].

0.0.4 Operations and finance

The *operations and finance* framework addresses infrastructure reliability and cost sustainability. Server-level metrics — *uptime*, *page load time*, *latency*, *server errors*, *client errors* — and compute-level metrics — *download speed*, *cpu/gpu usage*, *memory usage*, *client queue times*, *job run times* — are easy to capture but carry day-to-day variability from load, job size, and client-side network conditions, so trend monitoring is more informative than point-in-time values. For open-source projects, GitHub repository analytics provide an additional operational window: development velocity (commit frequency, pull-request review latency, merge frequency), repository-health composites, accumulating signals of technical debt (stale pull requests, abandoned branches, unresolved issues), and contributor dynamics that separate core-team commits from external community contributions and so register whether training and outreach are broadening adoption.

The Council of Councils identified sustainability as the "critical issue" for the CFDE, primarily because the programs that generate data are finite in duration [5]. *Funding cost* is the most accessible metric, but a complete picture decomposes it into *sample collection*, *sample measurement*, *QC*, *data storage*, *data preservation* (beyond the grant period), *computing hardware*, *server maintenance*, *cloud-based computing*, and *personnel*. The National Academies six-step framework for forecasting biomedical information-resource costs identifies cost-driver categories — data content, metadata, labor, infrastructure, compliance, governance — and notes that labor (curation and user support) consistently represents the single largest cost element [22]. Compliance costs in particular (HIPAA, FISMA, regulatory governance for human-subjects data such as Kids First and GTEx) are often underestimated in initial budgeting, and cloud-computing costs add provider-specific, usage-dependent volatility that resists multi-year forecasting [23].

The harder sustainability question is the break-even point for data sharing: mathematical models suggest aggregate research-community gains from reuse do not always outweigh the curation costs borne by data generators, which argues for selective investment in high-value datasets rather than uniform support. *Cost per data reuse event* — total curation and hosting cost divided by documented reuse instances — gives a concrete, comparable KPI for portfolio-level decisions, separating direct benefits (economies of scale in data generation and curation) from extended benefits (downstream scientific discovery and clinical outcomes enabled by reuse).

A complete catalog of metrics across the three frameworks, with collection difficulty, evaluation value, lifecycle phase, and interpretive notes, appears in Supplementary Table S1.

0.1 Implementation in practice

The CFDE Integration and Coordination Center (ICC) operates a portfolio-wide dashboard that demonstrates this evaluation model is implementable. The ICC currently collects a prioritized subset of the metrics described above (Table Table 1), drawing on NIH-provided structural data (NIH CFDE Website, NIH RePORTER) for funding and project context, PubMed and NIH iCite (including the Relative Citation Ratio) for publication and citation impact, GitHub for repository use, quality, and operations across the open-source software components, and Google Analytics for website engagement. Project teams onboard by labeling GitHub repositories with their NIH core project number (for example, "R01AG123456") and granting the ICC a Google Analytics service account, with NIH program staff supporting the trust and access conversation.

Table 1: Data sources and metrics collected by the CFDE ICC dashboard as of January 2026. This is the subset of the full catalog (Supplementary Table S1) selected for feasibility and informativeness across the portfolio.

Source	Metrics collected
NIH CFDE Website	Funding Opportunities, ID, Activity Codes
NIH RePORTER	Project name, funding amounts, project descriptions, project start/end dates, PI names and affiliations, project keywords
NIH RePORTER (self-reported)	Publications, presentations, grants
PubMed	Publication details, citations

Source	Metrics collected
NIH iCite	Citation details and metrics (e.g., Relative Citation Ratio)
GitHub	Repository use (stars, forks, etc.)
GitHub	Repository quality (README, LICENSE, etc.)
GitHub	Repository operations (commits, pull requests, issues, etc.)
GitHub	Repository narrative summaries via AI
Google Analytics	Website engagement metrics (page views, new vs. returning users, top countries, etc.)

The dashboard supports project- and portfolio-level views with ORCID-based authentication and role-based access, so teams see their own data while program managers see the full portfolio (Figure Figure 2). A cross-center Metrics Working Group, including representatives from CFDE centers and NIH program staff, reviews coverage and plans new metrics as the portfolio grows.

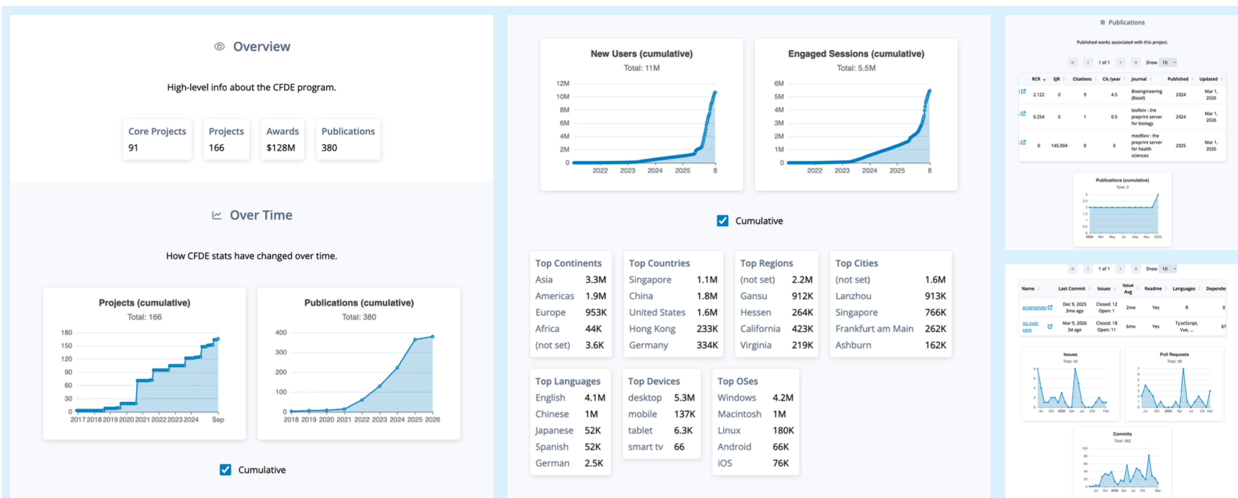

Figure 2: Screenshot of the CFDE ICC dashboard showing project- and portfolio-level metrics. Authentication and authorization controls follow CFDE community consensus on private versus public access.

Pilots underway target the gaps the frameworks identify. Full-text mining of accession numbers in publications addresses the data-citation gap directly. Event-organizer surveys, AI-generated narrative summaries of GitHub repositories, and structured-interview augmentation via generative AI [24] add the qualitative dimension that usage statistics cannot capture. What the implementation demonstrates is not that the metric set is final — it is that lifecycle-aware portfolio evaluation is operationally tractable at NIH scale, in near real time, with the burden on individual projects manageable.

0.2 Concluding remarks

No framework operates in isolation, and quantitative metrics alone cannot capture the full picture. The NIH Common Fund's "360-degree view" of program performance combines quantitative metrics with the contextual understanding only qualitative inquiry can provide [25]. Mixed-method designs — usage statistics and citation data alongside structured user interviews, focus groups, and surveys — are essential for understanding *why* resources succeed or fail, not just *what* is happening [26]. Two designs suit ecosystem evaluation: *sequential explanatory* (quantitative data first, then qualitative inquiry to explain observed patterns) and *convergent* (both collected simultaneously and triangulated at interpretation) [27]. The ICC dashboard already integrates the quantitative side; pilot user surveys and AI-augmented interviews are starting to add the qualitative.

Logic models that connect inputs (funding, data, personnel) to activities (harmonization, training, curation), outputs (portals, trained users, tools), and outcomes (publications, funded grants, practice change) [28] give evaluators the scaffolding needed to identify which elements work. For the CFDE, two questions matter: *to what extent are the elements of the logic model being implemented?* and *what are the critical paths through the model that lead to success?* [29]. These models should be living documents, revised as programs transition from pilot to full implementation.

The harder challenge is closing the "know-do gap" — the distance between the information metrics provide and actual decision-making. Evidence from healthcare quality improvement suggests that repeated, structured measurement itself drives behavioral change among data producers and managers, a Hawthorne effect [30]. Making metrics visible and regularly updated, as the ICC dashboard does, creates the sustained attention that incrementally improves data quality and FAIR compliance across the portfolio. NIH already requires regular FAIRness assessments and usage reporting [31], and the 2023 Data Management and Sharing Policy creates a structural opportunity to establish pre-policy baselines for sharing and reuse — a rare chance to test rigorously whether mandates translate into measurable ecosystem improvements.

The CFDE's Metrics Working Group plays the role that GA4GH's "Driver Projects" model has demonstrated elsewhere [32]: iterating on a small priority set of metrics with NIH program staff and cross-center representatives before expanding portfolio-wide. What this paper proposes is not a final metric set but a shared evaluation language for federated biomedical data ecosystems — one that is lifecycle-aware, mixed-method by design, and operationally tractable at NIH scale.

0.3 Acknowledgments

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0.4 Author contributions

All authors meet the International Committee of Medical Journal Editors (ICMJE) criteria for authorship. Ethan M. Lange, Vikram Adithya Ganesh, Faisal Alquaddoomi, David Mayer, Adriana Ivich, Vince Rubinetti, Casey Greene, and Sean Davis each made substantial contributions to the conception or design of the work and to the acquisition, analysis, or interpretation of data; participated in drafting the manuscript or revising it critically for important intellectual content; approved the final version for publication; and agreed to be accountable for all aspects of the work.

0.5 Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Claude from Anthropic in order to improve readability and clarity. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Outstanding Questions

- At what break-even point does the curation cost of a shared dataset get recovered by downstream reuse, and how should that point inform the decision to invest in further sharing versus letting a resource sunset?
- As the data-citation gap persists, what KPI should replace raw citation counts as the primary measure of data-resource impact, and how do we collect it at portfolio scale without burdening

individual projects?
- How should cross-disciplinary co-authorship arising from a shared dataset be weighted relative to direct citation when evaluating a resource's scientific influence?
- What is the right cadence for re-assessing the FAIR maturity of an active data resource, who owns the re-assessment, and how do we keep the process tractable as portfolios grow?
- As AI-driven and agent-based traffic on data portals continues to grow, how do we partition it into noise to remove versus signal to interpret as evidence of resource value?
- What governance structures keep federated evaluation frameworks honest as they evolve, so that metrics continue to serve funders' strategic questions rather than drifting toward data managers' reporting convenience?

Key References

1. Wilkinson MD et al. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 2016 [8]. The foundational definition of the principles that anchor the scientific-quality framework throughout this paper.
2. Wilkinson MD et al. A design framework and exemplar metrics for FAIRness. *Scientific Data*, 2018 [9]. Operationalizes FAIR as machine-measurable indicators, which is what makes automated FAIR-compliance monitoring tractable at the portfolio scale we describe.
3. Tedersoo L et al. Data sharing practices and data availability upon request differ across scientific disciplines [13]. Provides the bibliometric evidence that fewer than 30% of secondary analyses formally cite source data, the core empirical basis for our argument that citation-count KPIs systematically understate reuse.
4. The GTEx Consortium. The Genotype-Tissue Expression (GTEx) project. *Nature Genetics*, 2013 [2]. The paradigmatic example of a high-reuse public biomedical data resource (8,630 citations as of January 2026), used here to anchor the public-value framework with a concrete case.
5. Centers for Disease Control and Prevention. Standard Evaluation Framework for Large Health Care Data [17]. The 49-attribute fitness-for-purpose framework whose representativeness and linkability attributes are particularly load-bearing for federated ecosystems like the CFDE.

Glossary

CFDE (Common Fund Data Ecosystem). NIH cross-cutting initiative that harmonizes metadata and tooling across 19 Common Fund programs so researchers can find and reuse data across the portfolio.
ICC (Integration and Coordination Center). The CFDE center responsible for cross-portfolio coordination, evaluation, and the centralized metrics dashboard described here.
DRC, KC, TC, CWIC. The other four CFDE centers: Data Resource Center (data discovery), Knowledge Center (cross-cutting biological insight), Training Center (skills development), Cloud Workspace Implementation Center (compute infrastructure for in-place analysis).
FAIR (Findable, Accessible, Interoperable, Reusable). A widely adopted set of principles for making research data reliably reusable.
FAIRplus-DSM (FAIRplus Dataset Maturity model). A six-level maturity scale that operationalizes the FAIR principles for life-sciences data, from Level 0 (no identifiers, local storage) to Level 5 (enterprise lifecycle management).
RDA (Research Data Alliance). International community that classifies FAIR indicators by priority — Essential, Important, Useful — and provides a tiered roadmap for incremental improvement.
RICE, MoSCoW, MCDA. Product-management prioritization frameworks. RICE scores work by Reach × Impact × Confidence / Effort. MoSCoW labels items Must-have, Should-have, Could-have, Won't-have. MCDA (Multi-Criteria Decision Analysis) combines multiple weighted dimensions into a composite score.

ELIXIR. The European life-sciences research infrastructure, which publishes a ten-metric framework for evaluating bioinformatics software (version control, discoverability, automated builds, test data, documentation, and more).
GA4GH (Global Alliance for Genomics and Health). International standards organization whose "Driver Projects" model — small pilots that road-test new standards before portfolio-wide adoption — informs the CFDE's Metrics Working Group.
PIsCO (Performance Indicators framework for Collection of bioinformatics resource metrics). Server-side framework for automated collection and visualization of software-quality metrics, allowing software health to feed an evaluation dashboard directly.
RCR (Relative Citation Ratio). NIH iCite metric that benchmarks an article's citations against expectations for its field.
DMS Policy (NIH Data Management and Sharing Policy). The 2023 NIH policy requiring grantees to plan for data sharing, which establishes a pre-policy baseline against which sharing and reuse improvements can be measured.

References

1. **The Cancer Genome Atlas Pan-Cancer analysis project.**
, John N Weinstein, Eric A Collisson, Gordon B Mills, Kenna RMills Shaw, Brad A Ozenberger, Kyle Ellrott, Ilya Shmulevich, Chris Sander, Joshua M Stuart
Nature genetics (2013-10) https://www.ncbi.nlm.nih.gov/pubmed/24071849
DOI: 10.1038/ng.2764 · PMID: 24071849 · PMCID: PMC3919969

2. **The Genotype-Tissue Expression (GTEx) project**
John Lonsdale, Jeffrey Thomas, Mike Salvatore, Rebecca Phillips, Edmund Lo, Saboor Shad, Richard Hasz, Gary Walters, Fernando Garcia, Nancy Young, … Helen F Moore
Nature Genetics (2013-05-29) https://doi.org/gd5z68
DOI: 10.1038/ng.2653 · PMID: 23715323 · PMCID: PMC4010069

3. **The "All of Us" Research Program** *New England Journal of Medicine* (2019-08-15) https://doi.org/ggjdzj
DOI: 10.1056/nejmsr1809937 · PMID: 31412182

4. **Making Common Fund data more findable: catalyzing a data ecosystem**
Amanda L Charbonneau, Arthur Brady, Karl Czajkowski, Jain Aluvathingal, Saranya Canchi, Robert Carter, Kyle Chard, Daniel JB Clarke, Jonathan Crabtree, Heather H Creasy, … Owen White
GigaScience (2022) https://doi.org/gsxq6b
DOI: 10.1093/gigascience/giac105 · PMID: 36409836 · PMCID: PMC9677336

5. https://dpcpsi.nih.gov/sites/default/files/Day2-1225PM-Final-Report-CFDE-CoC-WG-HorwitzWilder_.pdf

6. **Decisions with multiple objectives: preferences and value tradeoffs**
Ralph L Keeney, Howard Raiffa
Wiley (1976)
ISBN: 9780471465102

7. **Advances in information security management & small systems security: IFIP TC11 WG11.1/WG11.2 Eighth Annual Working Conference on Information Security Management & Small Systems Security, September 27-28, 2001, Las Vegas, Nevada, USA**
Jan HP Eloff (editor)
Kluwer Academic Publishers (2001)
ISBN: 9780792375067

8. **The FAIR Guiding Principles for scientific data management and stewardship**
Mark D Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E Bourne, … Barend Mons
Scientific Data (2016-03-15) https://doi.org/bdd4
DOI: 10.1038/sdata.2016.18 · PMID: 26978244 · PMCID: PMC4792175

9. **A design framework and exemplar metrics for FAIRness**
Mark D Wilkinson, Susanna-Assunta Sansone, Erik Schultes, Peter Doorn, Luiz Olavo Bonino da Silva Santos, Michel Dumontier
Scientific Data (2018-06-26) https://doi.org/gfkrrt
DOI: 10.1038/sdata.2018.118 · PMID: 29944145 · PMCID: PMC6018520

10. **An analysis of the effects of sharing research data, code, and preprints on citations**
Giovanni Colavizza, Lauren Cadwallader, Marcel LaFlamme, Grégory Dozot, Stéphane Lecorney, Daniel Rappo, Iain Hrynaszkiewicz
PLOS ONE (2024-10-30) https://doi.org/g8p57w
DOI: 10.1371/journal.pone.0311493 · PMID: 39475849 · PMCID: PMC11524460

11. **Bibliometrics: The Leiden Manifesto for research metrics**
Diana Hicks, Paul Wouters, Ludo Waltman, Sarah de Rijcke, Ismael Rafols
Nature (2015-04-22) https://doi.org/gdg44f
DOI: 10.1038/520429a · PMID: 25903611

12. **Bibliometrics: tracking research impact by selecting the appropriate metrics**
Ashok Agarwal, Damayanthi Durairajanayagam, Sindhuja Tatagari, SandroC Esteves, Avi Harlev, Ralf Henkel, Shubhadeep Roychoudhury, Sheryl Homa, NicolásGarrido Puchalt, Ranjith Ramasamy, … Asher Bashiri
Asian Journal of Andrology (2016) https://doi.org/f877jx
DOI: 10.4103/1008-682x.171582 · PMID: 26806079

13. **Bibliometric analysis of neuroscience publications quantifies the impact of data sharing**
Herve Emissah, Bengt Ljungquist, Giorgio A Ascoli
openRxiv (2023-09-13) https://doi.org/hbsw6q
DOI: 10.1101/2023.09.12.557386 · PMID: 37745378 · PMCID: PMC10515804

14. **Medical Science Data Value Evaluation Model: Mixed Methods Study**
Dandan Wang, Yaning Liu
JMIR Medical Informatics (2025-08-21) https://doi.org/hbsw6s
DOI: 10.2196/63544 · PMID: 40839787 · PMCID: PMC12369987

15. **Big Data in Medicine Is Driving Big Changes**
K Verspoor, F Martin-Sanchez
Yearbook of Medical Informatics (2014-08) https://doi.org/gtwq2b
DOI: 10.15265/iy-2014-0020 · PMID: 25123716

16. **Data sharing, management, use, and reuse: Practices and perceptions of scientists worldwide**
Carol Tenopir, Natalie M Rice, Suzie Allard, Lynn Baird, Josh Borycz, Lisa Christian, Bruce Grant, Robert Olendorf, Robert J Sandusky
PLOS ONE (2020-03-11) https://doi.org/gnm52c
DOI: 10.1371/journal.pone.0229003 · PMID: 32160189 · PMCID: PMC7065823

17. **A Standard Framework for Evaluating Large Health Care Data and Related Resources**
Suad El Burai Felix, Hussain Yusuf, Matthew Ritchey, Sebastian Romano, Gonza Namulanda, Natalie Wilkins, Tegan K Boehmer
MMWR Supplements (2024-05-09) https://doi.org/hbsw6r
DOI: 10.15585/mmwr.su7303a1 · PMID: 38713639 · PMCID: PMC11078514

18. **Reproducibility of computational workflows is automated using continuous analysis**
Brett K Beaulieu-Jones, Casey S Greene
Nature Biotechnology (2017-03-13) https://doi.org/f9ttx6
DOI: 10.1038/nbt.3780 · PMID: 28288103 · PMCID: PMC6103790

19. **Bioconductor: open software development for computational biology and bioinformatics**
Robert C Gentleman, Vincent J Carey, Douglas M Bates, Ben Bolstad, Marcel Dettling, Sandrine Dudoit, Byron Ellis, Laurent Gautier, Yongchao Ge, Jeff Gentry, … Jianhua Zhang
Genome Biology (2004-09-15) https://doi.org/c2xm5v
DOI: 10.1186/gb-2004-5-10-r80 · PMID: 15461798 · PMCID: PMC545600

20. **Top 10 metrics for life science software good practices**
Haydee Artaza, Neil Chue Hong, Manuel Corpas, Angel Corpuz, Rob WW Hooft, Rafael C Jiménez, Brane Leskošek, Brett G Olivier, Jan Stourac, Radka Svobodová Vařeková, … Daniel Vaughan
F1000Research (2016-08-16) https://doi.org/gfvpq7
DOI: 10.12688/f1000research.9206.1 · PMID: 27635232 · PMCID: PMC5007752

21. **Indicators**
FAIRplus Data Maturity
https://fairplus.github.io/Data-Maturity/docs/Indicators/

22. **The Cost-Forecasting Framework: Identifying Cost Drivers in the Biomedical Data Life Cycle**
Engineering National Academies of Sciences, Policy and Global Affairs, Division on Earth and Life Studies, Division on Engineering and Physical Sciences, Board on Research Data and Information, Board on Life Sciences, Computer Science and Telecommunications Board, Committee on Applied and Theoretical Statistics, Board on Mathematical Sciences and Analytics, Committee on Forecasting Costs for Preserving and Promoting Access to Biomedical Data
Life-Cycle Decisions for Biomedical Data: The Challenge of Forecasting Costs (2020-06-08) https://www.ncbi.nlm.nih.gov/books/NBK562708/

23. **Migrating a research data warehouse to a public cloud: challenges and opportunities**
Michael G Kahn, Joyce Y Mui, Michael J Ames, Anoop K Yamsani, Nikita Pozdeyev, Nicholas Rafaels, Ian M Brooks
Journal of the American Medical Informatics Association (2021-12-17) https://doi.org/hbs2bg
DOI: 10.1093/jamia/ocab278 · PMID: 34919694 · PMCID: PMC8922165

24. **Harnessing the Power of AI in Qualitative Research: Role Assignment, Engagement, and User Perceptions of AI-Generated Follow-Up Questions in Semi-Structured Interviews**
He Zhang, Yueyan Liu, Xin Guan, Jie Cai, John M Carroll
arXiv (2025) https://doi.org/hbs2fv
DOI: 10.48550/arxiv.2509.12709

25. **Common Fund Data Ecosystem (CFDE) | NIH Common Fund** https://commonfund.nih.gov/dataecosystem

26. **Purposeful Sampling for Qualitative Data Collection and Analysis in Mixed Method Implementation Research**
Lawrence A Palinkas, Sarah M Horwitz, Carla A Green, Jennifer P Wisdom, Naihua Duan, Kimberly Hoagwood
Administration and Policy in Mental Health and Mental Health Services Research (2013-11-06) https://doi.org/f7nxzd
DOI: 10.1007/s10488-013-0528-y · PMID: 24193818 · PMCID: PMC4012002

27. **A process for conducting mixed methods data analysis**
John W Creswell, Machiko Inoue
Journal of General and Family Medicine (2024-10-03) https://doi.org/hbs2bf
DOI: 10.1002/jgf2.736 · PMID: 39776872 · PMCID: PMC11702478

28. https://www.wkkf.org/resource-directory/resources/2004/01/logic-model-development-guide

29. **Using logic model and visualization to conduct portfolio evaluation.**
Huang Wu, Jianping Shen, Jeffrey Jones, Xingyuan Gao, Yunzheng Zheng, Huilan Y Krenn
Evaluation and program planning (2019-02-26) https://www.ncbi.nlm.nih.gov/pubmed/30861395
DOI: 10.1016/j.evalprogplan.2019.02.011 · PMID: 30861395

30. **Changing health care provider performance through measurement**
Kenneth L Leonard, Melkiory C Masatu
Social Science & Medicine (2017-05) https://doi.org/f98fz3
DOI: 10.1016/j.socscimed.2017.03.041 · PMID: 28371629 · PMCID: PMC5616207

31. **NOT-OD-21-013: Final NIH Policy for Data Management and Sharing** https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-013.html

32. **Strategic Road Map** https://www.ga4gh.org/about-us/strategic-road-map/